

research_unit5(syllabus_based)

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User prompt: Pokhara University Faculty of Science and Technology Course No.: CMP 270 Full marks: 100 Course title: Research Fundamentals (2-0-2) Pass marks: 45 Nature of the course: Theory & Practical Total Lectures: 45 hrs Level: Bachelor Program: BE (Computer, IT and Software) 1. Course Description This course is designed to develop the skills of students to do a project/research work using the fundamental concepts of research. This course introduces what the research or project work is, explores in brief how it is done, explains what research ethics must be followed during the research/project work and finally guides the students how the research/project documentation is done in the form of the proposals report and final research/project report. 2. General Objectives • To acquaint the students with basic knowledge of research/project work. • To develop the skills in students to conduct research/project work. • To develop the skills in students to work in a team. • To develop the skills in students to write an impressive proposal report and final research/project report and present their work orally. • To acquaint the students with the knowledge of research ethics. 3. Methods of Instruction Lecture, Discussion and Project work. 4. Contents in Detail. Specific Objectives Contents • Understand the basic concepts of research, purpose and outcomes of a research/project work. Unit 1: Introduction (4 hrs) 1.1 What is research? 1.2 Research Aim and Objectives 1.3 Features of Research 1.4 Types of Research 1.5 The 6Ps of Research 1.6 Purpose of Research- reasons for doing research 1.7 Product of Research- outcomes of research 1.8 Research and Project • Understand and implement the research process model to conduct a research/project work. Unit 2: Research Process Model (10hrs) 2.1 Personal Experiences and Motivation 2.2 Literature Review 1. Purpose and objectives of a literature review 2. Literature resources 3. Conducting a literature review 4. Citation and its types 5. Bibliographic Detail and Referencing Systems 6. Plagiarism 2.3 Research Question 2.5 Conceptual framework 2.5 Strategies 1. Survey 2. Design and Creation 3. Experiment 4. Case Study 5. Action Research and 6. Ethnography 2.6 Data Generation Methods 1. Interview 2. Observations 3. Questionnaire 4. Documents 5. Types of triangulation in a research project 2.7 Data Analysis 1. Quantitative and 2. Qualitative data analysis • Familiarize with the laws and ethics in research conduction. Unit 3: Participants and Research Ethics (4hrs) 3.1 Participants 3.2 The law and Research 3.3 Rights of People Directly Involved 3.4 Responsibilities of an Ethical Researcher • Familiarize with the research proposal and its components. • Develop a research/project proposal. Unit 4: Proposal Writing (4hrs) 4.1 What is a research proposal? 4.2 Need of a Research Proposal 4.3 Components of a Research Proposal 4.4 A case study on any research paper/project • Familiarize with the research/project report and its components. • Develop a research/project report. Unit 5: Report Writing (4hr) 5.1 What is a research report? 5.2 Need of a Research Report 4.3 Components of a Research Report 4.4 A case study on any research paper/report 5. Practical Work Laboratory works of 30 hours per group of maximum 24 students should cover all the concepts of research fundamentals studied in the lectures. Students must find a new problem, write a proposal, solve the problem as their project/research work and submit a final project/research report and present their work orally. The marks for the practical work will be based entirely on their project/research work. The entire project/research work shall be divided into two phases and evaluation shall be done accordingly: Phase I: • The students are grouped in teams each containing at most 4 students. • Each team chooses a problem to solve as their project/research work and they work in a team. • They must define clearly what the problem is, justify why they choose the problem and how they will solve it and submit this as a proposal report (based on Unit 2 and 4). • Each team presents their proposal orally. Phase II: • After the approval of their proposal, they start working on the project. • Each team follows the research process studied in Unit 2 to do their project/research work. • Students keep reporting their progress about the project/research work to their instructor. • Students complete the project/research work, develop the

final project/research report (based on Unit 2 and 5) and again present it orally. 6. Evaluation system Internal Evaluation Weight Marks External Evaluation Marks Theory 30 Attendance & Class Participation 10% Assignments 20% Presentations/Quizzes 10% Semester-End examination 50 Internal Assessment 60% Practical (Project/research Work) 20 Proposal Report 30% Project Presentation 10% Final Project Report 40% Completeness of Project 20% Total Internal 50 Full Marks: 50 + 50 = 100 7. Student Responsibilities: Each student must secure at least 45% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course. 8. Prescribed Books and References 1. Oates, B. J., Griffiths, M., & McLean, R. (2022). *Researching information systems and computing*. Sage. 2. Walia, A. M., & Uppal, M. (2020). *Fundamentals of Research*. Notion Press. 8. Annex A. Components of Research Proposal 1. Title Page 2. Abstract 3. Table of Contents, List Figures, List of Tables and Abbreviations 4. Introduction a. Rationale/background b. Problem and motivation c. Aim and objectives of research d. Significance of research e. Scope of research f. Limitation 5. Literature review 6. Research problem and Solution 7. Methodology a. Research design b. Participants c. Data collection methods d. Data analysis techniques e. Ethical considerations f. Validation Techniques 8. Data Analysis and Findings 9. Discussions and Conclusion 10. Contributions and Future Works 11. Reference list/bibliography 12. Annexes B. Components of Research Report: 1. Title 2. Abstract 3. Keywords 4. Rationale/background and motivation 5. Aim and objectives of research 6. Literature review 7. Research problem 8. Methodology 9. Research plan and budget 10. Contributions-impact and significance 11. Reference list/bibliography 12. Annexes Other Parts of Research Report: 1. Funding and Acknowledgement 2. Table of Contents, List Figures, List of Tables and Abbreviations 3. Title Page and Copyright Page 4. Declaration and Recommendation 5. Certification Page Note: The components of research proposal and research report may vary based on the research strategy and nature of the research problem. 1. a) How do you define research? What are the specific features of research? Elaborate in detail. b) What are the types of research? Explain each with their characteristics and applications. c) What do you mean by research Methodology? Explain the good characteristics of research with example. d) How can you differentiate between applied research and basic research? Explain by the examples 2. a) What is the role of literature review? How do you organize the literature review? Explain. b) What are the sources of good literature? How can you better organize the literature review in project development? c) A bank wants to conduct research to evaluate the user experience of its mobile banking application. Suggest two suitable research strategies for this context and explain in brief. d) A government organization wants to conduct a research to find-out the rate of IoT device user in last five years in Nepal. Which method of research strategies do you prefer and why? 3. a) What do you mean by research strategy? Write down the difference between case study and ethnographic study with suitable examples. b) What are the different steps of research? Why does the c) What is Quantitative Research? What are the various data analysis techniques for Quantitative Research? d) What do you mean by the conceptual framework of a research process model? Discuss the importance of framework in research. 4. a) Who is an ethical researcher? Explain the responsibilities of an ethical researcher with examples. b) What do you mean by participant? Explain the rights of the participants. c) What are the rights of people directly involved in the research process? Explain. d) What are the responsibilities of Ethical researcher? 5. a) What is a research proposal? Explain the need of a research proposal. b) Explain about the components of research proposal. c) What are the responsibilities of Ethical researcher? d) What do you understand by data collection? Explain the various methods of data collection with example. 6. a) What are the components of a research report? Explain the importance of result and discussion in a report. b) Why do we need Research Report? How Research Findings are presented in Research Report? Explain in detail. c) What are the components of Research Proposal? How

objectives of a Research are formulated? Explain in detail. d) How does a well- written report help future researcher? Also discuss the importance of report writing in research. 7. Write short notes on: (Any two) a) Plagiarism b) IEEE style c) 6 P's of research d) ethical and unethical research e) ethical challenges in research . ii uploaded syllabus exam questions and note. now your job is to answer me question based on pdf(watch for depth) add your points as additional points(dont' keep answers limited to pdf) answer acguta question by yourself see syllabus as reference.

Response: Since you provided a comprehensive syllabus and a list of exam questions, I will start by thoroughly answering the first question from your list based on the provided text, while also expanding on it with additional academic context.

Question 1. a) How do you define research? What are the specific features of research? Elaborate in detail.

Definition of Research

Research is fundamentally defined as a systematic, problem-oriented, and truthful investigation. The primary aim of this investigation is to increase our understanding of a specific phenomenon, establish concrete facts, or discover new insights. It is best understood as an ongoing journey rather than a single event, functioning to translate abstract theoretical knowledge into practical, real-world applications.

Within the specific context of Computer Engineering, research takes on a highly applied nature. It involves inventing new computational algorithms, rigorously evaluating software architectures, optimizing network protocols, or designing novel Artificial Intelligence models to solve complex computational problems. Furthermore, all research is grounded in foundational philosophical worldviews, such as Positivism for objective metrics or Pragmatism for mixed-methods studies.

Specific Features of Research

According to the provided document, a credible scientific investigation must possess several defining characteristics:

- **Systematic (Structured & Logical):** Research cannot be conducted randomly or haphazardly. It requires a predefined, step-by-step roadmap to ensure flawless logic and to prevent the skipping of critical phases (for example, collecting data only after finalizing the system design).
- **Problem-Oriented (Purpose-Driven):** Scientific inquiry is never aimless; it always originates from a specific question, conflict, or gap in current knowledge. The entire methodology is built around identifying and solving this core problem in a measurable way.
- **Truthful/Objective (Unbiased):** High credibility requires that study findings are shaped purely by empirical facts and evidence. A researcher must entirely remove personal emotions, preconceived biases, or commercial interests from their data interpretation.
- **Empirical (Evidence-Based):** Valid research must rely strictly on concrete evidence rather than speculation, rumors, or unverified theories. Conclusions must be backed by measurable, real-world observations, hardware benchmarks, or software execution metrics.

- **Replicable (Verifiable & Repeatable):** A scientific breakthrough is only considered valid if it can be repeated. The methodology must be documented so clearly that an independent researcher can achieve identical results using the same setup and data collection rules.

Additional Expert Perspectives (Beyond the Syllabus)

While the document provides a robust foundational definition, modern academic research also demands a few additional characteristics:

- **Cyclical and Iterative:** Research rarely ends with a definitive answer. Finding a solution to one problem typically exposes new variables and generates new questions, creating a continuous, iterative cycle of knowledge-building.
- **Rigorous:** The procedures applied to find answers must be heavily scrutinized to ensure they are relevant, appropriate, and strictly justified, minimizing the margin for statistical or logical error.
- **Peer-Scrutinized:** In the modern academic ecosystem, research is not considered fully complete until it has survived the critical scrutiny of independent experts in the field (peer review). This ensures the methods and conclusions hold up against universal scientific standards before being accepted as fact.

Would you like me to tackle question **1. b) (Types of research)** next, or is there another specific question from the exam list you would prefer to prioritize?

User prompt: 5.1 What is a research report?

Response: 5.1 What is a Research Report?

Based on the provided document, a research report is defined as a formal, highly structured document that clearly details the systematic process, methodology, and final outcomes of an empirical investigation.

Core Characteristics from the Text:

- **Official Record:** Rather than simply presenting subjective opinions, it acts as the official, objective record of a scientific inquiry.
- **Translating Complexity:** It takes raw, complex field data, statistical analyses, and laboratory metrics and translates them into a clear, readable narrative.
- **Comprehensive Narrative:** The report explains exactly what specific problem was studied, the methods used to investigate it, what the resulting data revealed, and the broader implications of those findings for the field.

Additional Expert Insights (Beyond the Syllabus):

To expand on this definition, a robust research report also serves several other critical functions in the scientific community:

- **Audience Customization:** While maintaining strict scientific rigor, the tone and emphasis of a research report often shift depending on the target audience. Reports aimed at academic peers heavily emphasize methodological limits and theoretical frameworks, whereas technical reports intended for corporate stakeholders or policymakers prioritize executive summaries and actionable, real-world recommendations.
 - **Archival and Cumulative Value:** A research report acts as a permanent intellectual time capsule. By securely archiving the exact datasets and coding structures used, it allows future researchers to conduct broad meta-analyses, preventing the duplication of effort and forming the foundation for future innovation.
 - **Standardized Global Communication:** To ensure universal readability, research reports strictly adhere to standardized formatting frameworks, most notably the **IMRAD** structure (Introduction, Methods, Results, and Discussion). This global standardization allows scientists from different countries and disciplines to quickly scan, navigate, and evaluate the validity of the work without getting lost in disorganized narratives.
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User prompt: 5.2 Need of a Research Report

Response: 5.2 Need of a Research Report

Based on the **Final BE 4 th sem(2)(1).pdf** document, a research report is a vital necessity for several structural and scientific reasons:

- **Knowledge Dissemination:** It takes raw, unprocessed data out of field notebooks or laboratory logs and formally shares it with the wider scientific community, policymakers, and local practitioners.
- **Scientific Accountability and Replication:** A clear, transparent report allows peer reviewers and independent researchers to scrutinize the exact methods used. This transparency is necessary so others can verify the results and repeat the experiment under similar conditions to test its true validity.
- **Evidence-Based Decision Making:** Policymakers and funding agencies rely heavily on formal reports to allocate resources, design community development programs, or draft strict regulations based on hard, empirical evidence rather than intuition or guesswork.
- **Preserving an Institutional Record:** It provides a permanent, easily accessible archive of historical data. This ensures that future researchers do not waste valuable time, energy, or institutional funds repeating studies that have already been conclusively resolved.

Additional Expert Insights (Beyond the Syllabus)

To expand on why research reports are strictly necessary, here are a few additional points that drive the academic and industrial ecosystems:

- **Evaluating Researcher Competence:** In academic environments, the report is the primary metric used by university committees to evaluate a student's or investigator's technical

capabilities, critical thinking skills, and mastery of the subject matter for grading or degree awarding.

- **Securing Future Funding:** Funding bodies and grant agencies use the final reports of completed projects to gauge the return on their investment. A well-documented, successful report builds trust and drastically increases the likelihood of securing financial backing for future investigations.
 - **Commercial and Industrial Application:** In fields like Computer Engineering, the research report often serves as the technical foundation for translating abstract theories into marketable patents, commercial software tools, or hardware prototypes.
 - **Highlighting New Gaps:** A good report doesn't just answer the original question; it actively identifies new anomalies and limitations encountered during the study, explicitly pointing future researchers toward the next logical questions that need to be answered.
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User prompt: 4.3 Components of a Research Report

Response: Based on the **Final BE 4 th sem(2)(1).pdf** document, the components of a research report (detailed under Section 5.3 in the text) are systematically organized into three distinct, sequential sections.

(Note: While the syllabus outline lists this as 4.3 under Unit 5, the detailed text correctly categorizes it as 5.3).

1. Preliminary Pages

- **Title Page:** Features a concise, descriptive title of the study, author names, institutional affiliations, and the date of submission.
- **Abstract:** A dense, single-paragraph summary (usually 150-250 words) that outlines the research problem, methodology, primary results, and core conclusions.
- **Table of Contents/Lists:** Visual guides that map out page numbers for chapters, sections, data tables, and figures.

2. Main Body (The IMRAD Format)

- **Introduction:** Establishes the background context, states the specific research problem, defines the study's scope, and outlines the primary objectives or hypotheses.
- **Literature Review:** A critical synthesis of existing academic work that identifies current research gaps and demonstrates how the new study builds upon prior knowledge.
- **Methodology:** A transparent blueprint detailing the target and accessible populations, sampling techniques, data collection instruments (like survey questionnaires), experimental designs, and analytical methods.
- **Results:** The objective presentation of the collected data using systematic tables, charts, or histograms, accompanied by straightforward textual explanations without subjective

interpretation.

- **Discussion:** The analytical heart of the report where the author interprets the findings, compares them to previous literature, addresses unexpected anomalies, and acknowledges study limitations.
- **Conclusion and Recommendations:** Summarizes the main takeaways and offers actionable, practical suggestions for field implementation, policy adjustment, or future research directions.

3. Supplementary Material

- **References / Bibliography:** A meticulously formatted list (e.g., in APA style) providing full academic credit for all external literature, books, and reports cited throughout the text.
- **Appendices:** Supplementary items that are too bulky for the main body but essential for full transparency, such as original blank survey questionnaires, raw mathematical derivations, or extensive statistical tables.

Additional Expert Insights (Beyond the Syllabus)

To make a research report truly comprehensive and publication-ready, several other elements are typically included in professional and academic settings:

- **Keywords:** Positioned immediately below the Abstract, this is a short list of 4-6 highly relevant terms. Keywords act as search tags, helping digital indexing databases (like IEEE Xplore or Google Scholar) correctly categorize the paper so other researchers can find it.
- **Acknowledgements:** Usually placed in the preliminary pages, this section formally thanks funding bodies, institutional review boards, supervisors, and community stakeholders who provided financial, technical, or administrative support.
- **Declaration of Originality (and Copyright Page):** Most universities require a formally signed page confirming that the document is the author's original work and has not been plagiarized or submitted for a different degree.
- **Limitations vs. Delimitations:** While the "Discussion" section covers limitations (weaknesses outside the researcher's control, like a low response rate), expert reports also explicitly define *delimitations* in the Introduction. Delimitations are the conscious, intentional boundary choices made by the researcher (e.g., intentionally excluding rural data to focus purely on urban networks).

User prompt: 4.4 A case study on any research paper/report

Response: Based on the **Final BE 4 th sem(2)(1).pdf** document, the practical application of report components is demonstrated through a specific case study (detailed under Section 5.4 in the text, though numbered 4.4 in the syllabus outline).

The text examines a mixed-methods research paper titled "**The Operational Impact and Student Perceptions of Generative AI Tools in Academic Collaborative Environments**" to show how

structural components function in practice.

Here is how the report was broken down structurally:

1. Title and Abstract (The Entryway)

- **The Title:** Clearly specified the independent variable (Generative AI tools) and the dependent context (student collaboration and performance).
- **The Abstract:** A dense 200-word paragraph that summarized the problem, the sample size (14 interviewees), the methodology (thematic synthesis), and the final institutional recommendations.

2. Introduction and Problem Statement (The Rationale)

- **The Problem:** Highlighted a specific research gap: while individual AI use is documented, its impact on team-based assignments and peer-to-peer collaboration is not.
- **Objectives:** Aimed to categorize the tasks students delegate to AI during group work and evaluate the impact on learning outcomes.

3. Literature Review (Mapping the Landscape)

- **Synthesis:** Synthesized past peer-reviewed work on computer-supported collaborative learning (CSCL) and human-AI interaction.
- **Identifying Gaps:** Used a literature matrix to contrast different findings, exposing a clash on whether AI enhances critical thinking or promotes dependency and plagiarism.

4. Methodology (The Operational Blueprint)

- **Parameters:** Mapped the target population (undergraduate engineering/management students) and accessible population (students at a specific mid-sized college).
- **Instruments:** Utilized a mixed-methods design, combining a standardized descriptive questionnaire with in-depth, semi-structured interviews of 14 participants.

5. Data Analysis and Results (The Empirical Evidence)

- **Quantitative Visualization:** Used tables and histograms to display descriptive survey metrics regarding AI usage frequency.
- **Qualitative Coding:** Processed raw interview text through a strict 5-step sequence, reducing text into codes mapping how students use AI (e.g., technical coding support, assignment assistance, team task optimization).

6. Discussion (The Analytical Interpretation)

- **Interpretation:** Linked field results back to macro-theories, analyzing the tension between AI speeding up operational tasks and the ethical risks of paraphrasing plagiarism or reduced critical thinking.

- **Validation:** Wove verbatim participant quotes directly into the text to support the insights.

7. Conclusion and Recommendations (The Impact)

- **Actionable Policy Output:** Concluded that outright AI bans are impractical. Instead, recommended a governance framework including mandatory AI literacy modules, clear boundary guidelines, and updated assessment methods like oral defenses.

8. Supplementary Infrastructure (References and Appendices)

- **Formatting and Transparency:** Included a complete APA-style reference list, alongside appendices containing the blank survey and interview guide so future investigators can replicate the study.

Additional Expert Insights (Beyond the Syllabus)

To expand on why analyzing a case study of a research paper is so critical for students:

- **Reverse-Engineering Structure:** Reading a finalized report acts as a reverse-engineering exercise. It helps novice researchers see how abstract concepts (like "triangulation" or "conceptual frameworks") actually look when translated into professional academic writing.
 - **Understanding Real-World Constraints:** Case studies highlight how researchers handle imperfect, real-world data. They show how authors pivot when sample sizes are smaller than expected or when qualitative data reveals unexpected themes outside the original hypothesis.
 - **Bridging Theory and Practice:** A strong case study demonstrates that a research paper is not just an academic hurdle, but a practical tool. As seen in the AI study above, the end goal is often actionable policy—proving that good research directly influences how institutions, businesses, and governments operate.
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